🌐 Module 05: Web Application Penetration Testing

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# Exercise 3: Performing Web Vulnerability Scanning using AI
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Scenario

In this lab, we will use AI tools such as ChatGPT to generate a Python script for automating web vulnerability scanning of a target application. The script will identify user input entry points, scan for known vulnerabilities such as SQL injection, XSS, and misconfigurations, and perform fuzz testing to analyze server responses for potential exploits.

Objective

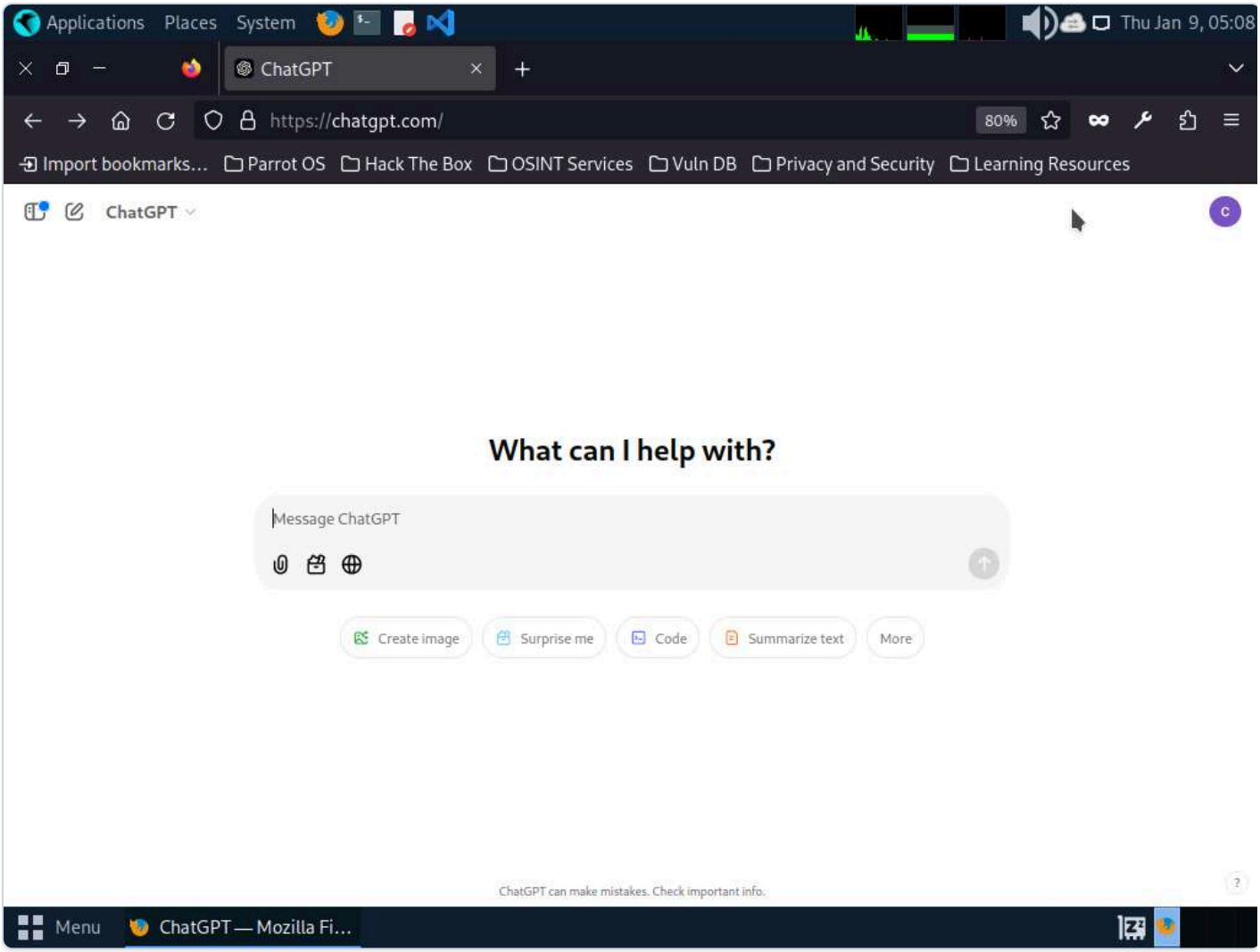
The objective of this lab is to help students learn how to leverage AI tools to generate custom automation scripts for web application entry-point identification, vulnerability scanning, and fuzzing.

Task 1: Performing Web Vulnerability Scanning using AI

In this task, we generate and execute a Python script for automated web vulnerability scanning against the vulnerable target web application `http://testphp.vulnweb.com/`.

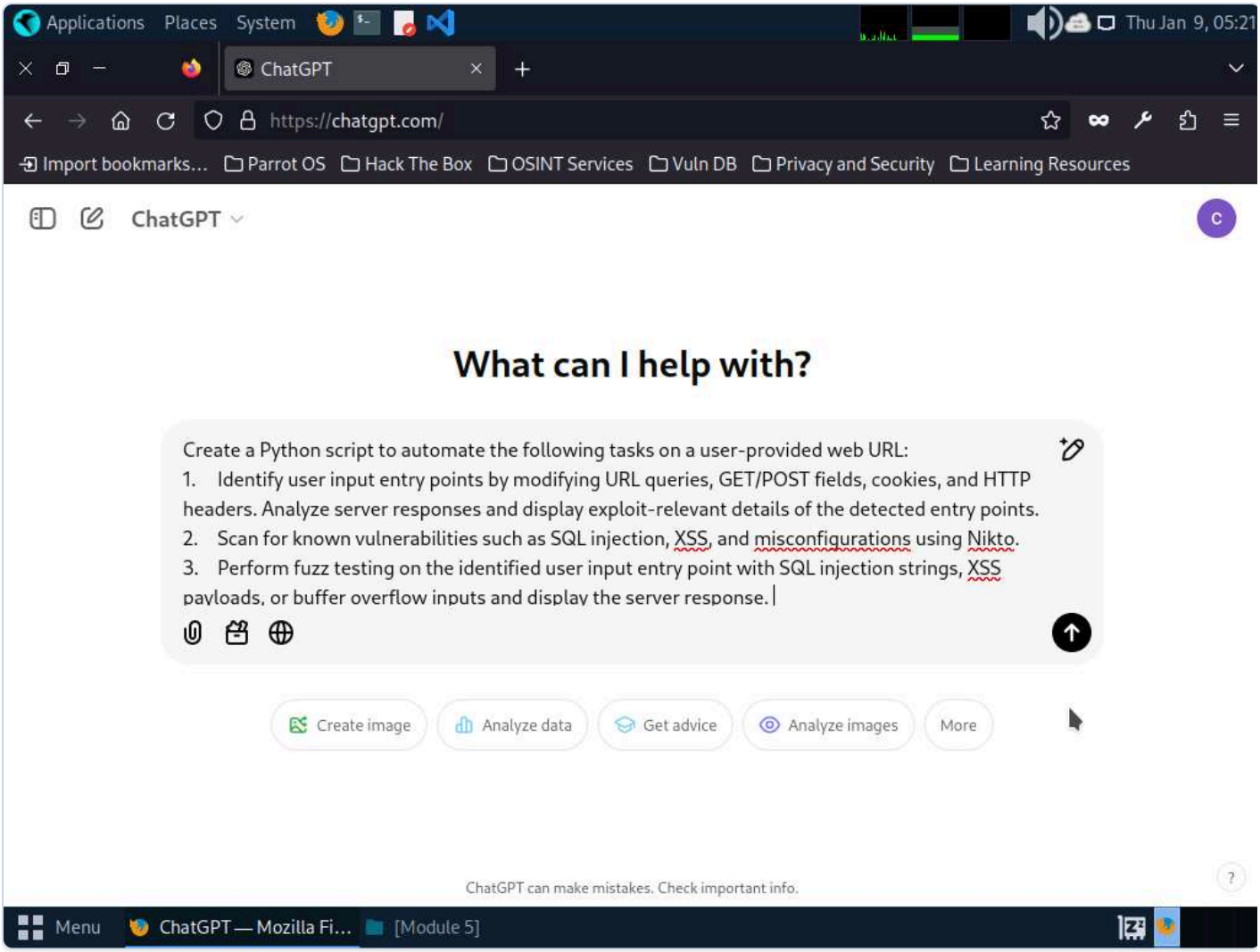
Lab Tasks

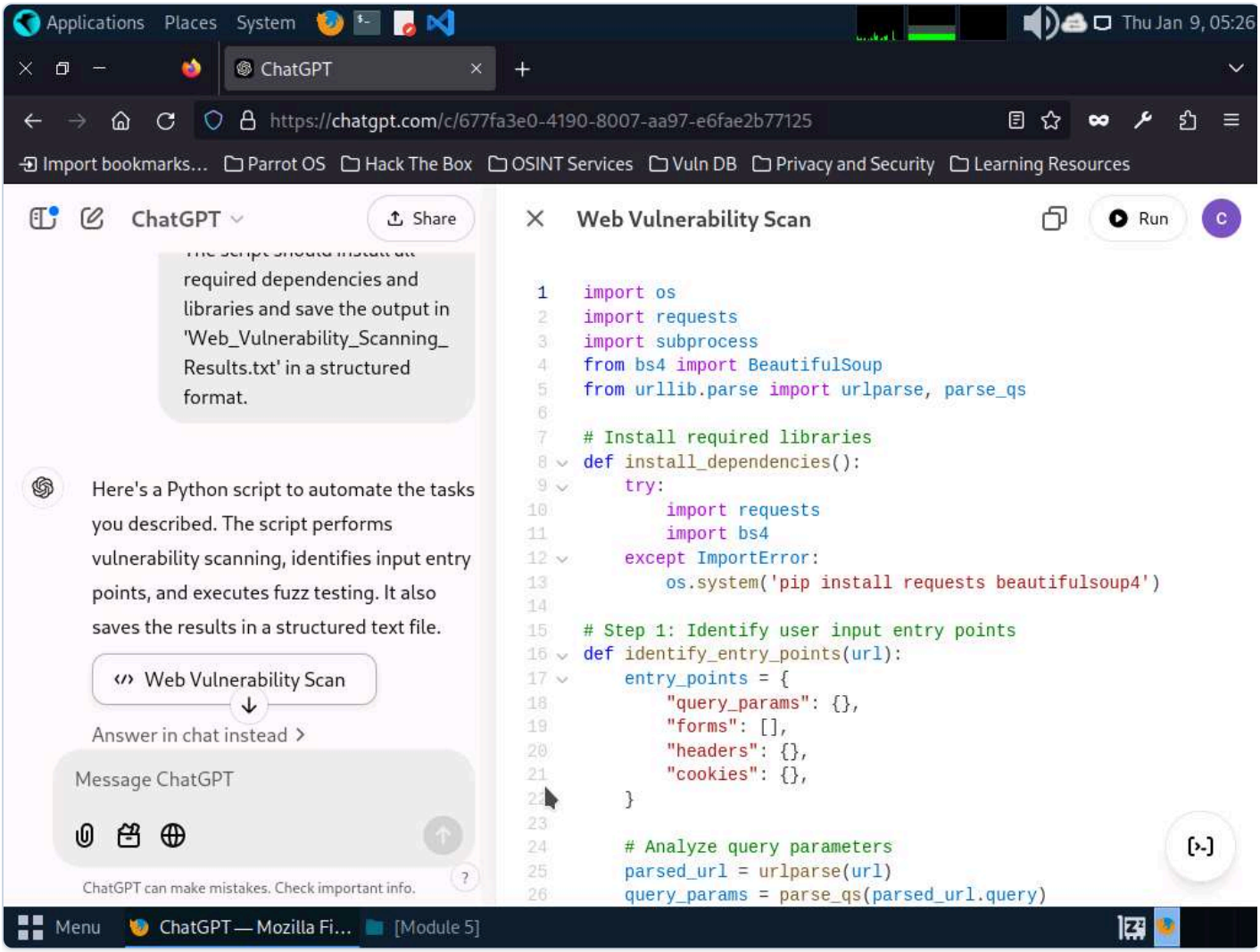
1. In the **Parrot Security** machine, open **Mozilla Firefox**, type **https://chatgpt.com/**, and press Enter.



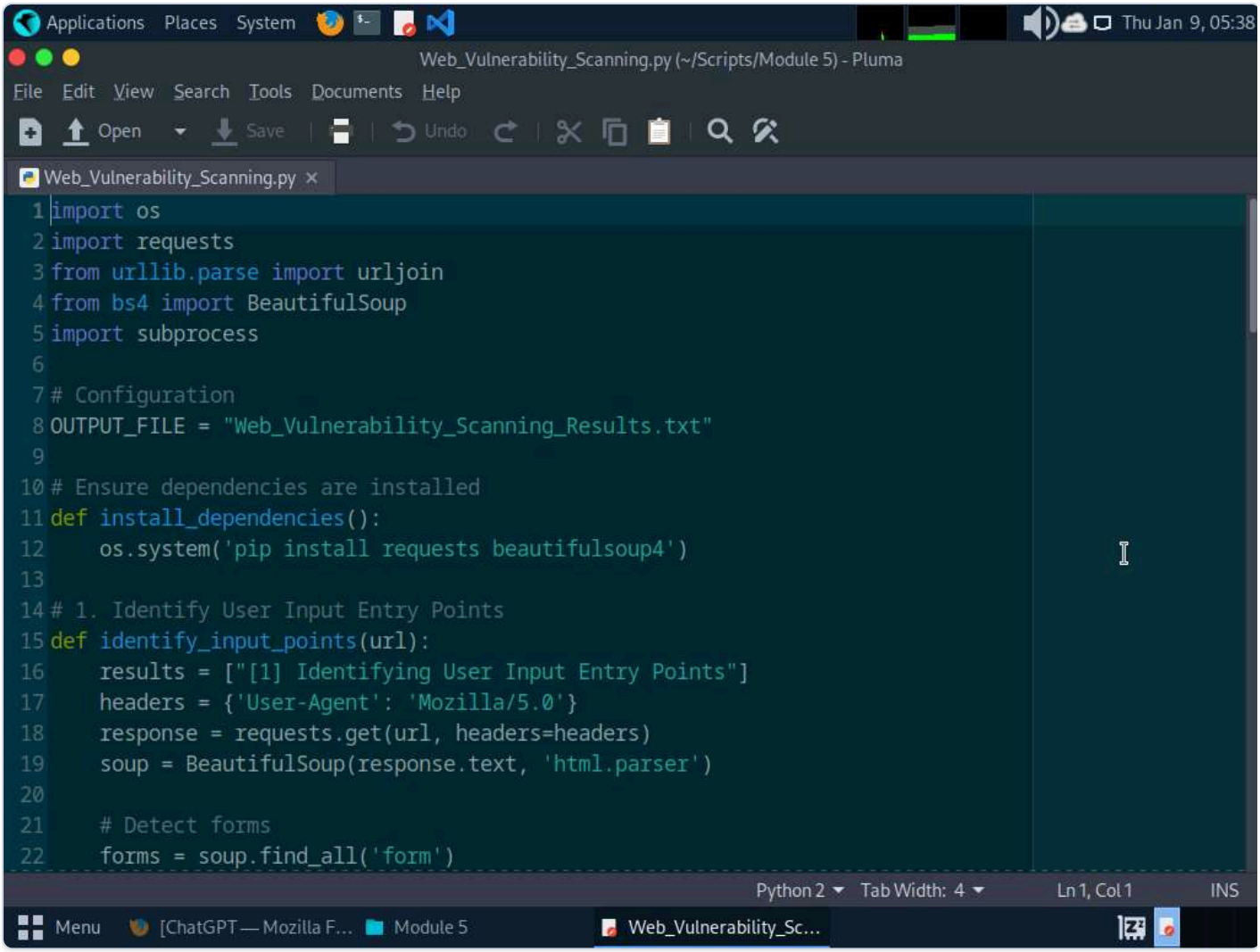
1. Log in to ChatGPT using your Gmail account if you are not logged in by clicking on the **Continue with Google** button.
2. In the Message ChatGPT text box enter the below sample prompt and press Enter or click on the Up arrow button for generating a Python script to automate web vulnerability scanning:

```
Create a Python script to automate the following tasks on a user-provided web URL:
1. Identify user input entry points by modifying URL queries, GET/POST fields, cookies, and HTTP he
2. Scan for known vulnerabilities such as SQL injection, XSS, and misconfigurations using Nikto.
3. Perform fuzz testing on the identified user input entry point with SQL injection strings, XSS pa
The script should install all required dependencies and libraries and save the output in 'Web_Vulne
```

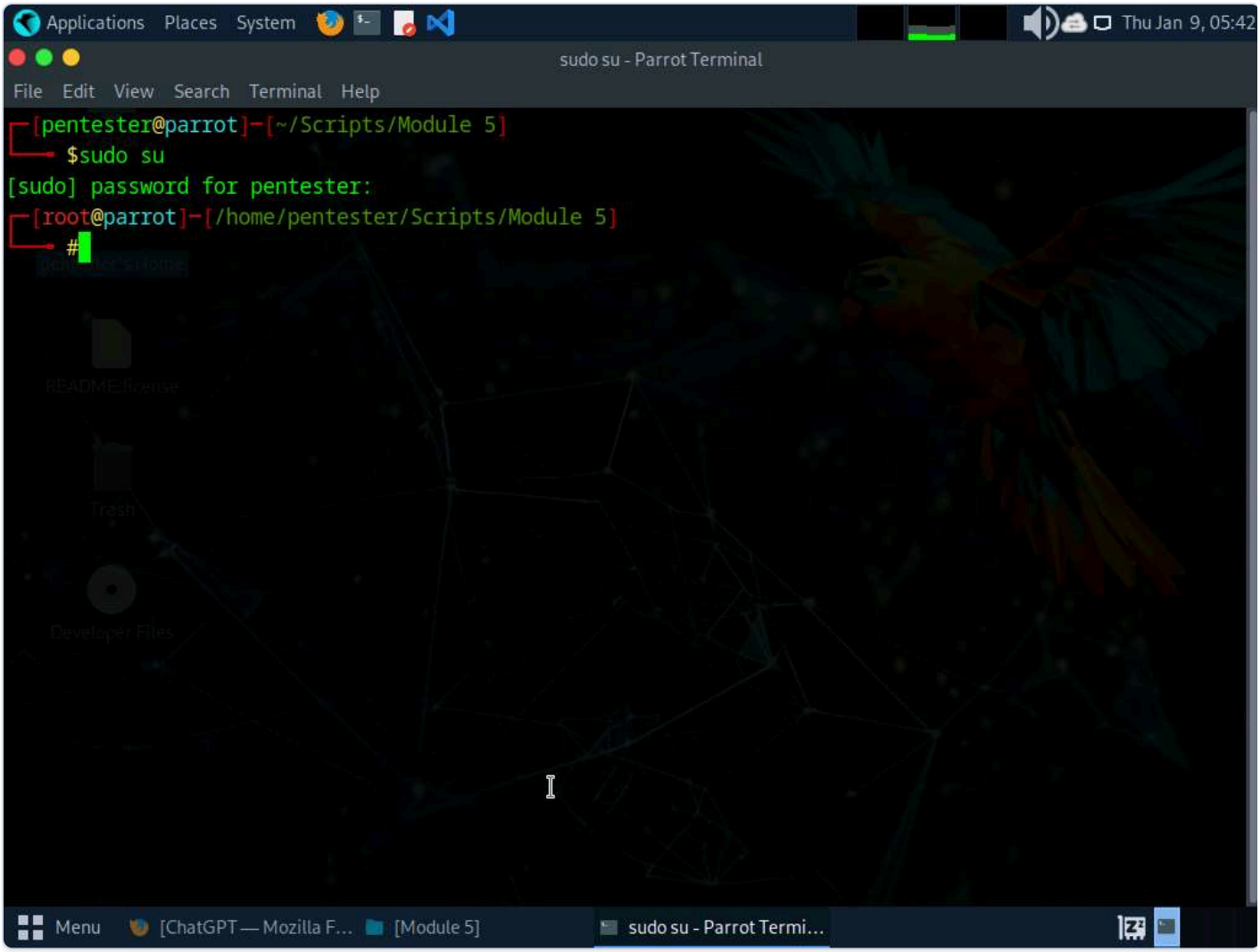

Applications Places System
Thu Jan 9, 05:21
ChatGPT
https://chatgpt.com/
Import bookmarks...
Parrot OS
Hack The Box
OSINT Services
Vuln DB
Privacy and Security
Learning Resources
ChatGPT
C
What can I help with?
Create a Python script to automate the following tasks on a user-provided web URL:
1. Identify user input entry points by modifying URL queries, GET/POST fields, cookies, and HTTP headers. Analyze server responses and display exploit-relevant details of the detected entry points.
2. Scan for known vulnerabilities such as SQL injection, XSS, and misconfigurations using Nikto.
3. Perform fuzz testing on the identified user input entry point with SQL injection strings, XSS payloads, or buffer overflow inputs and display the server response.
Create image
Analyze data
Get advice
Analyze images
More
ChatGPT can make mistakes. Check important info.
Menu
ChatGPT — Mozilla Fi...
[Module 5]



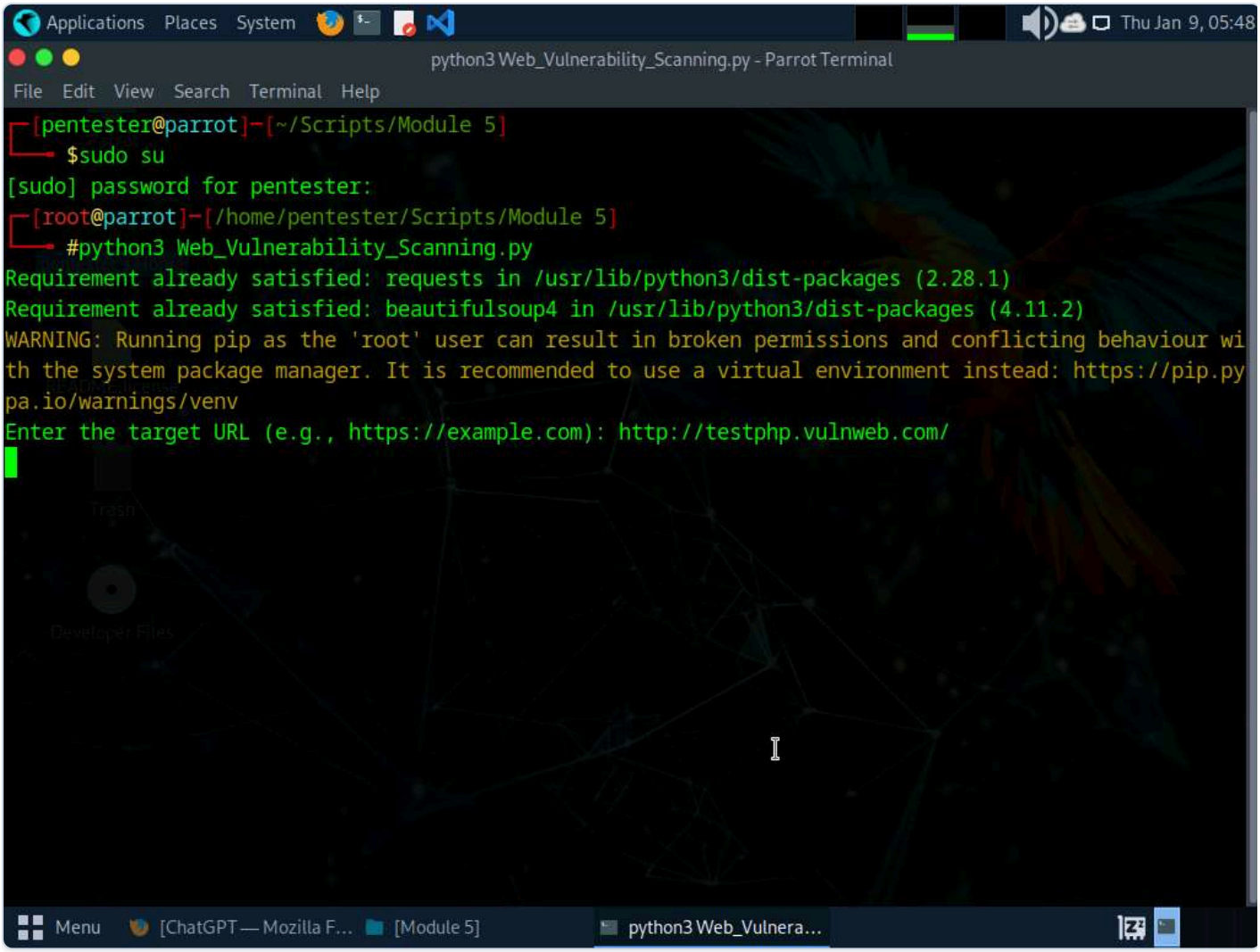
- *(You can get different outputs when performing this lab.)*
- Now, copy and save the generated Python Script in a text file named **Web_Vulnerability_Scanning.py**.
- You can run the sample Python script that is already saved and available in your local repository at: `CPENTv2 Pen Testing Resources/CPENTv2 Scripts/Module 05 Web Application Penetration Testing/Web_Vulnerability_Scanning.py`

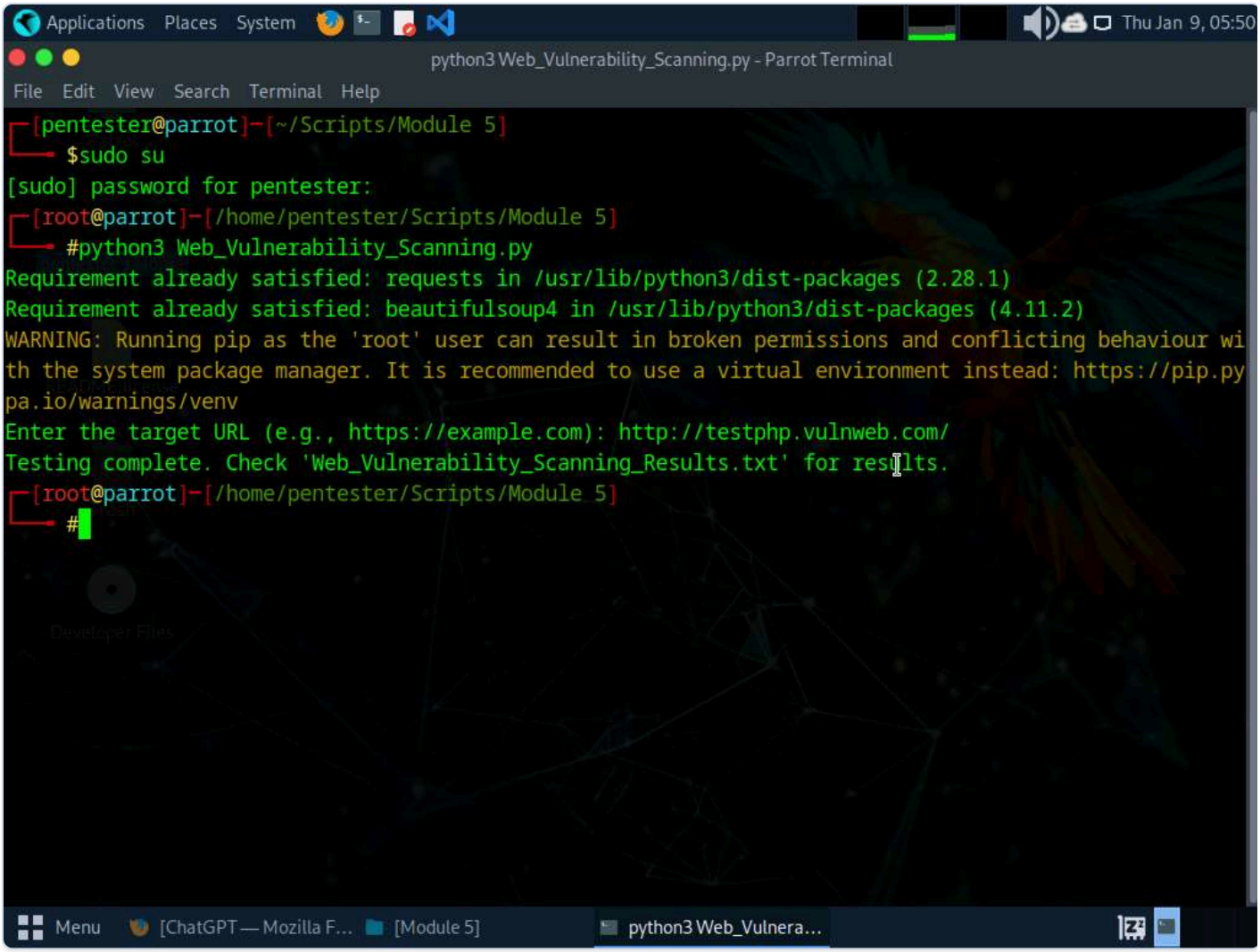


1. Launch the command-line terminal, type **sudo su** and press Enter. Type **toor** as password.

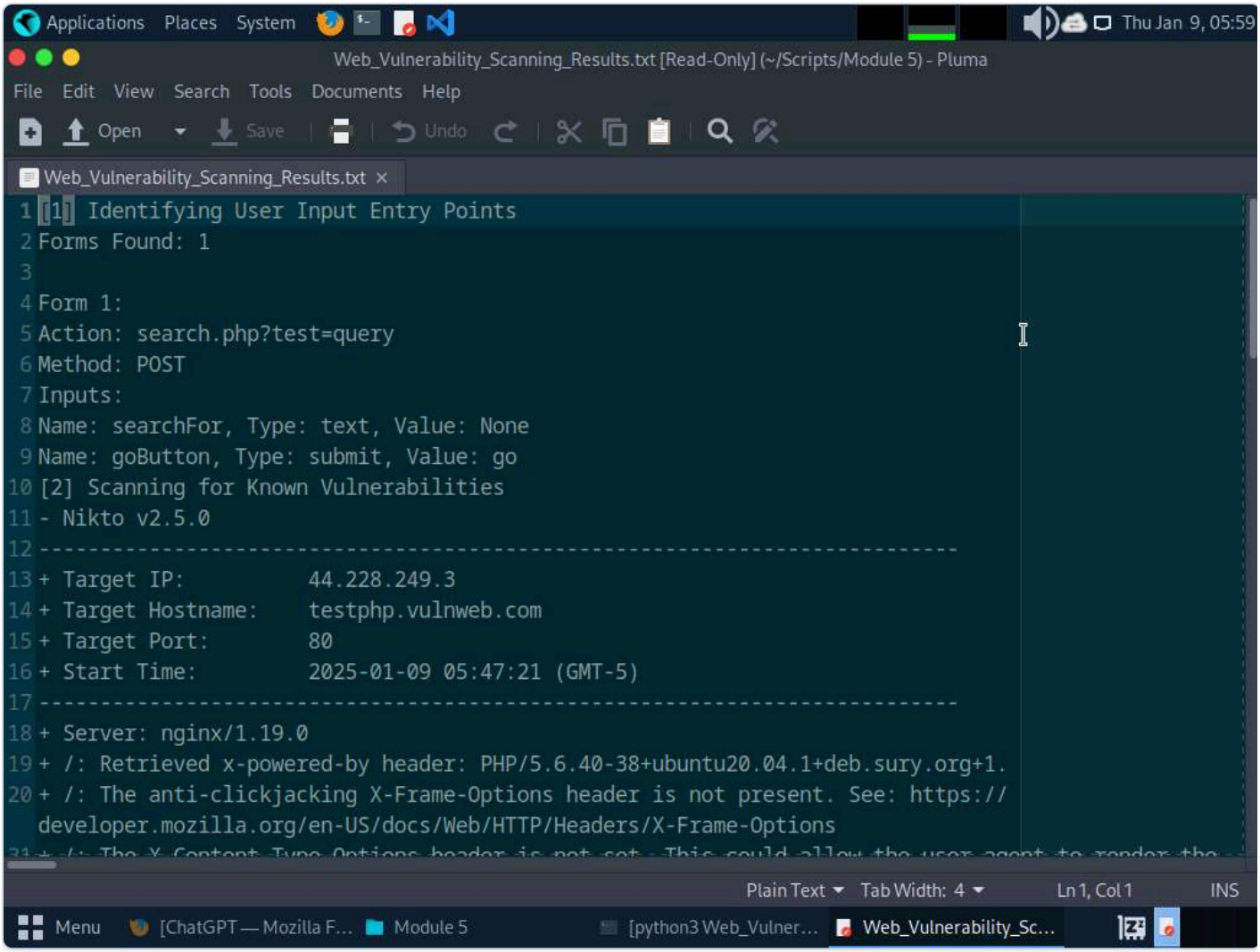


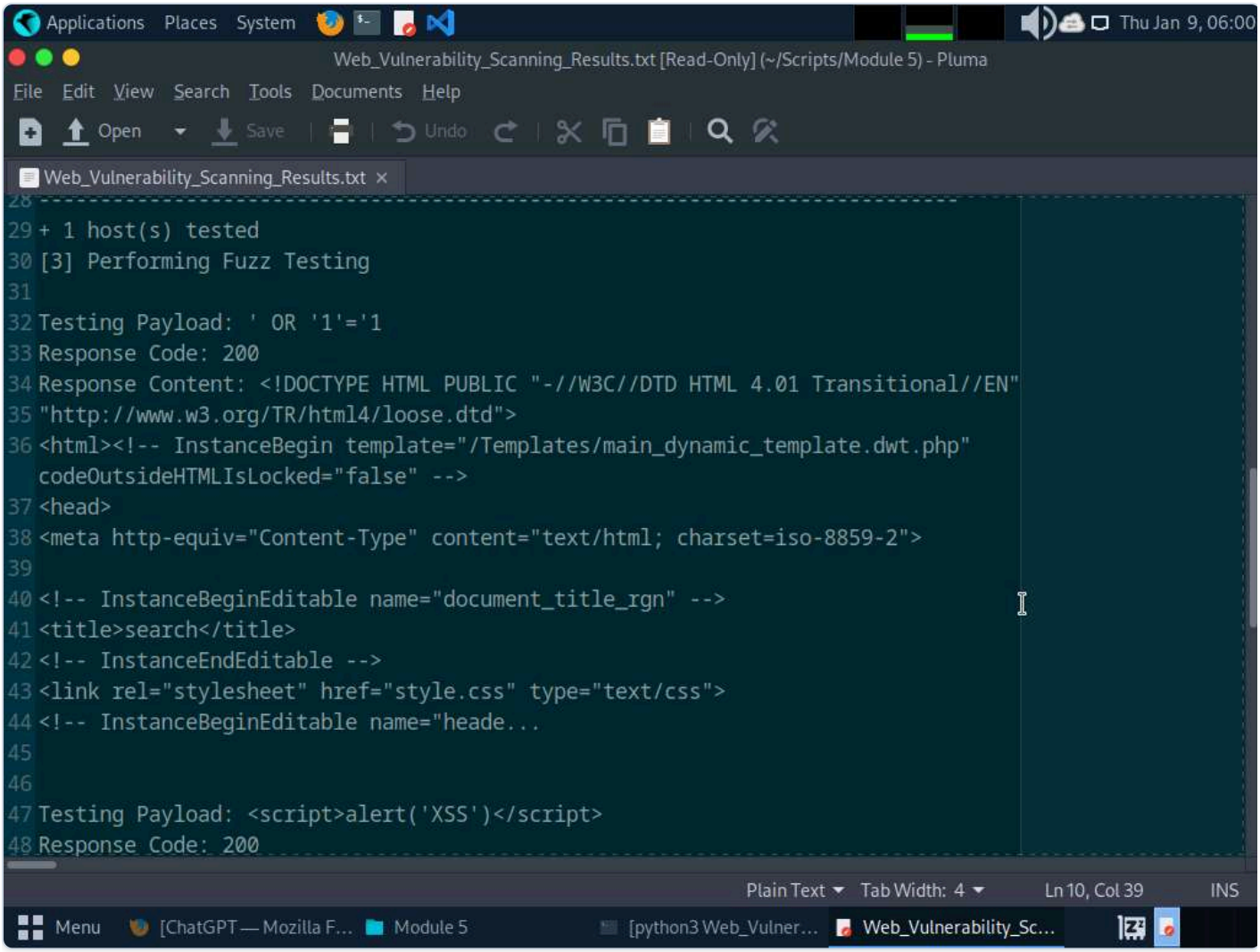
1. Go to the saved script location in the terminal, type **python3 Web_Vulnerability_Scanning.py**, and press Enter. Once executed, it will ask you to enter the target URL.
2. Enter **http://testphp.vulnweb.com/** as the target URL.


Applications Places System
Thu Jan 9, 05:48
python3 Web_Vulnerability_Scanning.py - Parrot Terminal
File Edit View Search Terminal Help
[pentester@parrot]-[~/Scripts/Module 5]
$sudo su
[sudo] password for pentester:
[root@parrot]-[/home/pentester/Scripts/Module 5]
#python3 Web_Vulnerability_Scanning.py
Requirement already satisfied: requests in /usr/lib/python3/dist-packages (2.28.1)
Requirement already satisfied: beautifulsoup4 in /usr/lib/python3/dist-packages (4.11.2)
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour wi
th the system package manager. It is recommended to use a virtual environment instead: https://pip.py
pa.io/warnings/venv
Enter the target URL (e.g., https://example.com): http://testphp.vulnweb.com/
Menu
[ChatGPT — Mozilla F...
[Module 5]
python3 Web_Vulnera...



1. After completion of the script execution, it will save the results in a separate file. Open the **Web_Vulnerability_Scanning_Results.txt** file to check for the obtained results.


Applications Places System
Thu Jan 9, 05:59
Web_Vulnerability_Scanning_Results.txt [Read-Only] (~/Scripts/Module 5) - Pluma
File Edit View Search Tools Documents Help
Open
Save
Undo
Web_Vulnerability_Scanning_Results.txt
[1] Identifying User Input Entry Points
Forms Found: 1
Form 1:
Action: search.php?test=query
Method: POST
Inputs:
Name: searchFor, Type: text, Value: None
Name: goButton, Type: submit, Value: go
[2] Scanning for Known Vulnerabilities
- Nikto v2.5.0
+ Target IP: 44.228.249.3
+ Target Hostname: testphp.vulnweb.com
+ Target Port: 80
+ Start Time: 2025-01-09 05:47:21 (GMT-5)
+ Server: nginx/1.19.0
+ /: Retrieved x-powered-by header: PHP/5.6.40-38+ubuntu20.04.1+deb.sury.org+1.
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://
developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
Plain Text
Tab Width: 4
Ln 1, Col 1
INS
Menu
[ChatGPT — Mozilla F...
Module 5
[python3 Web_Vulner...
Web_Vulnerability_Sc...



1. Analyze the results for:
2. User Input Entry Points
3. Known Vulnerabilities
4. Fuzz Testing

This concludes the lab exercise. Close all open windows.

> **IMPORTANT:** ❓ **Question 5.3.1.1** On the Parrot Security machine, utilize AI tools, such as ChatGPT, to generate a Python script for automating web vulnerability scanning of `http://testphp.vulnweb.com/`. Enter the IP Address of `http://testphp.vulnweb.com`. [!SUCCESS] ✅ **Answer:** **44.228.249.3**